Summer Internship Project Specification
AI-Powered Campaign Builder

Duration: 2 Months (8 Weeks)

Technology Stack: React, ASP.NET Core Web API, SQL Server, OpenAI/Azure OpenAI

Project Overview

The objective of this internship is to design and develop a marketing campaign management platform that allows users to build customer selections, create Email and SMS campaigns, schedule campaign delivery, and automatically generate complete campaigns using an AI Agent.

Functional Requirements

1. Customer Selection Builder

Create audience selections using configurable filters such as age, gender, country, registration date, and loyalty points. Selections must be saved and reusable.

2. Email Campaign Builder

Create email campaigns with subject lines, rich-text or HTML content, preview functionality, and draft management.

3. SMS Campaign Builder

Create SMS campaigns with content validation, character counting, and preview capabilities.

4. Campaign Scheduler

Schedule campaigns for future delivery and manage campaign lifecycle states such as Draft, Scheduled, Sent, and Cancelled.

5. Dashboard

Display campaign statistics, upcoming scheduled campaigns, and overall campaign activity.

AI Agent (Advanced Phase)

The AI Agent must be capable of generating a complete campaign from a simple business request. It should automatically create audience criteria, email content, SMS content, and campaign scheduling information.

Suggested Database Entities

Campaign, Selection, EmailTemplate, SmsTemplate, and Schedule.

Project Timeline

Week 1: Requirements analysis, architecture design, database design, and wireframes.

Week 2: Backend foundation and database implementation.

Week 3: Frontend foundation and API integration.

Week 4: Customer Selection Builder implementation.

Week 5: Email and SMS Campaign Builders implementation.

Week 6: Campaign Scheduler and workflow implementation.

Week 7: AI Agent integration and campaign generation.

Week 8: Testing, documentation, final presentation, and demo.

Expected Deliverables

• Source Code (Frontend, Backend, Database)
• Database Schema and Scripts
• Technical Documentation
• User Guide
• Final Demonstration
• AI Campaign Generation Feature